

Byeongjee Kang

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RESEARCH INTERESTS

Programming Languages, Compiler, Formal Methods, Parallelism, and WebAssembly

EDUCATION

Carnegie Mellon University

Pittsburgh, USA

Ph.D. in Software Engineering, advised by Prof. Limin Jia

Aug 2023 – Present

Pohang University of Science and Technology (POSTECH)

Pohang, Korea

Bachelor of Computer Science and Engineering/Mathematics (double major)

Feb 2018 – Feb 2023

- GPA : 4.26/4.3 (ranked 1st in class)
- Entered with *Program for highly Dedicated Students (PHDs)*, presidential scholarship and supporting program offered to one outstanding high school student in the year
- Graduated within 6 semesters (leave of absence for military duty during Jan 2020 – Dec 2021)

Korea Science Academy of KAIST (*High School for Science-Gifted Students*)

Busan, Korea

Graduated

Feb 2015 – Feb 2018

- GPA : 4.23/4.3 (ranked 1st in class)
- Funded by Korea Science Academy Fund Scholarship

PUBLICATION

GrFeyn: Efficient Parallel Sparse Simulation of Quantum Circuits, Sam Westrick, Pengyu Liu, [Byeongjee Kang](#), Colin McDonald, Mike Rainey, Mingkuan Xu, Jatin Arora, Yongshan Ding and Umut Acar, IEEE International Conference on Quantum Computing and Engineering (QCE) 2024 (Best Paper) (pdf)

Narrowing and Heuristic Search for Symbolic Reachability Analysis of Concurrent Object-Oriented Systems, [Byeongjee Kang](#) and Kyungmin Bae, Science of Computer Programming 2024 (doi.org/10.1016/j.scico.2024.103097)

Symbolic Reachability Analysis of Distributed Systems using Narrowing and Heuristic Search, [Byeongjee Kang](#) and Kyungmin Bae, International Workshop on Formal Techniques for Safety-Critical Systems (FTSCS) 2022 (doi.acm.org?doi=3563822.3568017)

PUBLICATION (UNDER REVIEW)

Fray: An Efficient General-Purpose Concurrency Testing Platform for the JVM, Ao Li, [Byeongjee Kang](#), Vasudev Vikram, Isabella Laybourn, Samvid Dharanikota, Shrey Tiwari, and Rohan Padhye, International Conference on Software Engineering (ICSE) 2025

PREPRINTS

Mr3: An Execution Framework for Hive and Spark Sungwoo Park, Seunggon Namgung, [Byeongjee Kang](#), Journal of Distributed and Parallel Computing (dx.doi.org/10.2139/ssrn.4587845)

HONORS AND SCHOLARSHIPS

ILJU Fellowship (ILJU Academy and Culture Foundation)

Aug 2023 – July 2028

- Received total of USD 12,000 over eight semesters during graduate studies

Program for Highly Dedicated Students (POSTECH)

Feb 2018 – Feb 2023

- Presidential scholarship and supporting program in POSTECH offered to **one outstanding freshman** each year

Presidential Science Scholarship (Ministry of Education of Korea)*Feb 2018 – Feb 2023*

- National science scholarship offered to top freshmen students in the field of science and technology

Global Leadership Program (CSE Department, POSTECH)*Sep 2019 – Present*

- Scholarship and research support program for selected students in computer science department

Domun Scholarship (POSTECH)*Feb 2019*

- Scholarship for freshmen with highest GPA

WORK EXPERIENCES**Kodebox**

Seoul, Korea

*Software engineer as military service alternative**Jan 2020 – Dec 2021*

- **CodeChain Foundry:** Open-source blockchain engine based on composable module system
 - Worked as software engineer
 - Implemented BLS signature aggregation scheme to optimize block seal size
 - Designed interface for modules to communicate with consensus engine
 - CodeChain Foundry is written in Rust
- **Zuzu:** Web-based captable management software
 - Worked as full-stack software engineer
 - Main features implemented include regression testing framework for document generation service, payment, and more
 - Tech Stacks: Django (Python), React (TypeScript), GraphQL, Kubernetes, Google Cloud Platform

TEACHING EXPERIENCES**Student Mentoring Program** (CSE Department, POSTECH)*Feb 2022 – June 2022*

- Served as tutor to teach Algorithm course

Intensive Research Program (POSTECH)*Jan 2019*

- Served as mentor in research experience program for science high school students

Voluntary Teaching (POSTECH Volunteering Club)*Jan 2019 – Dec 2019, Sep 2022 – Dec 2022*

- Served as voluntary teacher of math for unprivileged middle school students

EXCHANGE STUDENT & LANGUAGE PROGRAM**Summer Session Program**

TU Berlin

*Data science course**Aug 2019 — Sep 2019***English Program**

University of New South Wales

*English course**Dec 2017 — Jan 2018***AwesomeMath Summer Program**

Cornell University

*Mathematics problem solving program for gifted middle and high school students**June 2016 — July 2016*

POSTECH Decentralized Autonomous Organization (PDAO) <i>Student association for blockchain development and research</i> • Served as mentor in blockchain core development team • Designed consensus algorithm to be used in PDAO	POSTECH <i>Feb 2022 – Present</i>
Intensive Research Program <i>Research experience program for science high school students - advised by Prof. Jae-Joon Kim</i> • Learned basic deep learning concepts and researched on techniques to reduce size of DNN	POSTECH <i>July 2017</i>
Research and Education (R&E) Program <i>Group research program for juniors - Advised by Dr. Youngchul Jeon</i> • Researched properties of <i>Abelian Ring</i>, a field of noncommutative ring theory • Presented result as school representative in Japan Super Science Fair (JSSF) 2016, at Ritsumeikan High School	Korea Science Academy of KAIST <i>Mar 2016 – Dec 2016</i>
Korea Young Physicists' Tournament (KYPT) <i>Team-based scientific competition for high school students</i> • Designed and performed experiments, and debated with other teams on the outcome • Won gold medal	The Korean Physical Society <i>June 2017 – Jan 2018</i>
POSCAT <i>Students club for algorithm study and competitive programming</i> • Committee member of Postech Programming Contest 2019	POSTECH <i>Feb 2018 – Dec 2019</i>
POSTECH Student Ambassador <i>Student-run POSTECH International PR group</i> • Voluntary service to support exchange students visiting POSTECH	International Relations, POSTECH <i>Feb 2018 – Dec 2018</i>
POBBA <i>Basketball club</i> • Served as club secretary during the year 2019	POSTECH <i>Feb 2018 – Present</i>

LANGUAGES

Korean (native), English (fluent)